

4.4 Sound Cards

Sound cards are expansion cards with chips to process audio information. With modern motherboards carrying such circuitry onboard, most users do not feel the need for a dedicated sound card. Many onboard sound processors are capable of supporting even 8-channel surround sound. Sound cards are used by audio-philosophers not satisfied by onboard sound quality, and professionals who find the features offered by onboard sound limited.

4.4.1 ADC / DAC

The Analogue-to-Digital Converter (ADC) is responsible for converting the analogue signals received through the sound source into digital signals capable of being processed by the PC. The reverse process of converting digital signals into analogue signals to be passed on to the speakers is done by the Digital-to-Analogue Converter. The quality of the conversion is influenced by the Sampling rate and Bit rate.

4.4.2 Bit Resolution

This refers to the amount of data that is used to store information about every sample while sampling (converting from one format to another). The better the bit resolution, the greater the fidelity of the recorded sound. Bit resolution is usually expressed in bits.

4.4.3 Bit Rate

The product of the sampling rate and the bit resolution is the bit rate. For example, if a conversion process takes 44.1 kHz as the sampling rate and 8 bits as the bit resolution per channel, the bit rate for the converted sound in mono would be 44.1 Kilobytes per second, and stereo, 88.2 Kilobytes per second.

4.4.4 Codec

A Codec, short for Compressor-Decompressor, is an algorithm used to convert and play back audio in a specific format. Audio stored by a particular codec can only be played back if the codec exists on